Southern Cross University

Psychotropic medications (CNS pharmacology)

Dr Ken Wojtkowski

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Overall objectives for psychotropic medications

- For each of the disorders listed below, briefly explain the clinical manifestations of the disorder, the hypothesised neurologic abnormalities, and the mechanisms by which the commonly prescribed medications help correct the neurologic abnormality.
  - Parkinson's disease
  - Alzheimer's disease
  - Anxiety disorders,
  - Depression,
  - Schizophrenia and other psychotic disorders and
  - Seizure syndromes
- And for each of the medications discussed, note any associated suffix, and explain the side effects and contraindications

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Medications that treat Neurodegenerative disorders: Parkinson's disease

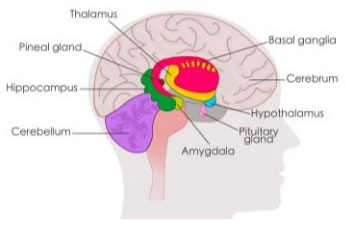
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Objectives for Part 1: Neurodegenerative Medicines

- Describe Parkinson's and Alzheimer's disease with respect to the neurologic changes that occur.
- Briefly explain the mechanism by which the following drugs exert their therapeutic effects, and the major indications, contraindications and side effects.
  - Parkinson's: carbidopa/levodopa (Sinemet); *Benzatropine (Benztropine)*
  - Alzheimer's: *donepezil*; *memantine*

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Parkinson's disease results in a decrease in dopamine-neurons in the basal ganglia



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Parkinson's disease

Affects part of the extrapyramidal system

- **The part of brain involved is basal ganglia**
  - **Controls involuntary movements**
  - **Initiation of muscular movement**
  - **Fine muscle movement**
  - **Muscle tone**
- **Characterised by pill-rolling tremor, hypokinesia, shuffling gate, muscle rigidity.**



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Parkinson's disease

Affects part of the _____ system

- **The part of brain involved is _____, which has these actions:**
  1. ___
  2. ___
  3. ___
  4. ___
- **Characterised by:**



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Parkinson's disease

- **Balance between cholinergic and dopaminergic nerve activity is upset by too little dopamine**
  - **Too much cholinergic activity relative to dopamine**
- **Two solutions:**
  - **Increase dopamine**
    - **Carbidopa/Levodopa**
  - **Decrease cholinergic activity**
    - **Antimuscarinics**
      - ***Benztropine***

We start with this:

Acetylcholine **Dopamine**



Parkinson's results in this:

Acetylcholine **Dopamine**



We compromise with this:

Acetylcholine **Dopamine**



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Antimuscarinic / anticholinergic medications

These drugs restore the balance between acetylcholine and dopamine (the ultimate result is that both are decreased)

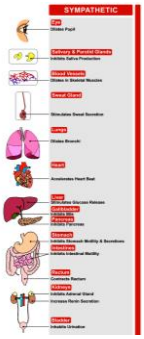
- Only useful in the initial stages and when the dopaminergic drugs begin to be ineffective
- Recall the side effects of drugs that **decrease** the parasympathetic (**rest and digest**) system
  - Dry mouth, blurred vision, difficulty in micturition,
  - Confusion and
  - Hallucinations
- Example: ***Benzatropine* (benztropine in US texts)**

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Benzatropine

Recall that the anticholinergic drugs will make the sympathetic nervous system dominate:

Dry mouth, constipation,
blurred vision,
difficulty in micturition, tachycardia



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Increasing dopamine

- Dopamine cannot cross the blood–brain barrier.
- **Levodopa**, or L-dopa (precursor to dopamine) can cross the BBB
  - Converted into dopamine in brain

BUT too much levodopa causes peripheral side effects

- The enzyme that converts levodopa to dopamine is a DOPA decarboxylase.
- **Carbidopa inhibits this enzyme**

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Carbidopa and Levodopa (Sinemet)

- **Many common adverse effects**
  - Nausea, depression, agitation and confusion (although these are sometimes due to the disease itself), increased sexual activity, hypotension, delusions, dysrhythmias, as well as involuntary movements of the extremities, head, lips and tongue (dyskinesias).
- On-off phenomenon
  - Characterised by the affected person enjoying near-normal movement until severe disease symptoms suddenly develop.

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Newer medications

All of the following aim to increase dopamine

(no need to learn)

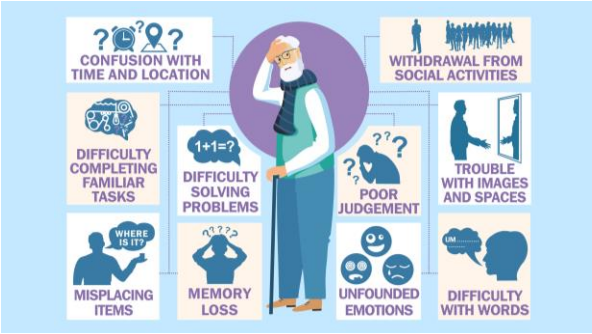
- **Selegiline, rasagiline and entacapone inhibit the breakdown of the enzymes that break down dopamine**
- **Amantadine inhibits the reuptake of dopamine from the synaptic cleft back into the first neuron**
- **Bromocriptine, pramipexole and ropinirole are dopamine agonists that can cross into the brain**

None of the medications reverse the changes that occur in this chronic progressive disorder

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Medications that treat Neurodegenerative disorders: Alzheimer's disease

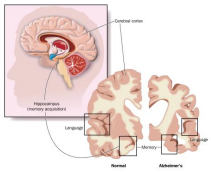
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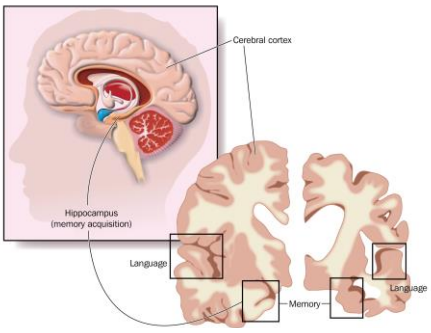
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Alzheimer's disease

- Alzheimer's disease is a form of dementia characterised by profound brain shrinkage
  - Progressive deterioration of intellectual and cognitive functioning.
- Two observations that have led to the current therapy
  - Acetylcholine is reduced early in the course of the disease
  - Glutamate levels have been found to be **raised**
- The acetylcholinesterase inhibitors are used to enhance cholinergic transmission, while the use of the glutamate receptor antagonist **memantine** reduces the excitotoxic effects of glutamate.



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Acetylcholinesterase inhibitors (*donepezil*)

By blocking the enzyme that degrades acetylcholine, more neurotransmitter will be available and its action prolonged.

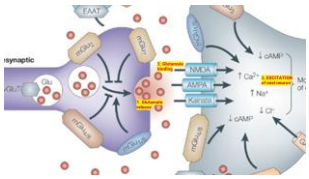
- **Common adverse effects**
  - Donepezil produces a number of common adverse effects, which include nausea, vomiting, elevated liver enzyme levels, diarrhoea, dizziness, anorexia, myalgia and dyspepsia

These drugs may reduce deterioration in cognition but do not stop the pathological process or yield a cure

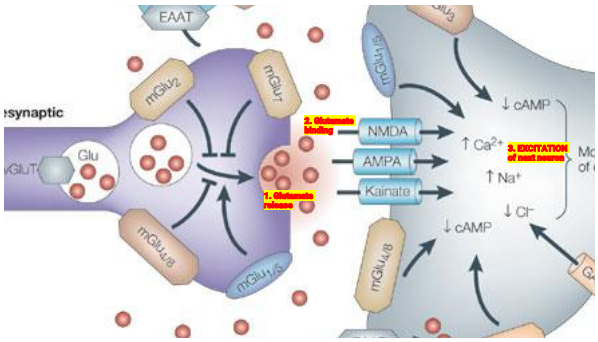
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Glutamate

- Glutamate is the most abundant excitatory neurotransmitter in the nervous system
- Glutamate binding results in increased Na^+, Ca^{2+} or decreased Cl^-, depending on the receptor to which it binds. (In each case, that leads to excitation.)
- For instance, glutamate binds to the NMDA receptor, allowing more calcium to flow in (more excitation)



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Memantine blocks glutamate at the NMDA receptor

- The brain is very vulnerable to glutamate-mediated overexcitation; this results in excitotoxicity.
- Ultimately, with this overexcitation, cell integrity is disrupted and the nerve cell dies.
- Part of the process of Alzheimer's disease may have to do with this excitotoxicity from excess glutamate.
- The psychotropic agent **memantine is a** selective NMDA receptor antagonist.
  - This decreases calcium influx and decreases excitation

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Glutamate receptor antagonists

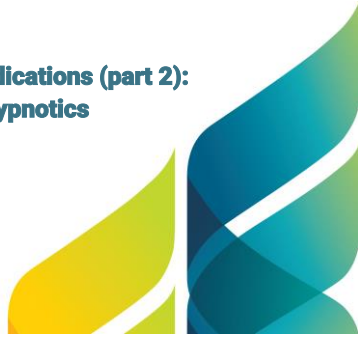
Memantine is indicated in the management of moderate to severe Alzheimer's disease.

- **Mechanism of action**
  - Memantine is an NMDA receptor antagonist that selectively restores the ungated receptors to the gated state and inhibits prolonged calcium ion influx without affecting normal glutamate neurotransmission.
- **Common adverse effects**
  - Agitation, insomnia, drowsiness, dizziness, headache, gastrointestinal disturbances and coughing.

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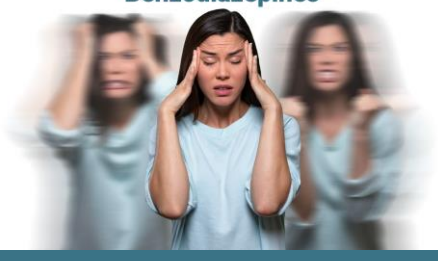
Psychotropic medications (part 2): Anxiolytics and hypnotics



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Anxiolytics and hypnotics
Benzodiazepines



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Anxiolytics and hypnotics

- Anxiolytics, minor tranquiliser and sedative are often used interchangeably and are described as a substance that diminishes the activity of the CNS. Sedatives calm a person receiving therapy, and moderate excitability.
- 'Hypnotic' is the term used to describe a substance that induces sleep
  - Often difficult to distinguish
- Many of the medicines in these categories can induce psychological dependence

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Objectives for Part 2
Anxiolytics and Hypnotics

For each of the anxiolytics listed below, give an example of the drug class, the general mechanism by which it exerts its therapeutic effects, the indications, contraindications and most important side effects.

- Benzodiazepines (*Diazepam* , *Alprazolam*, *Clonazepam* and *Lorazepam*)
- Azapirones (*Buspirone*)

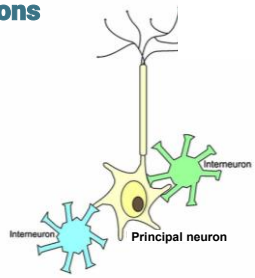
Make sure you know the ZEPAM and ZOLAM SUFFIXES!

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Interneurons vs neurons

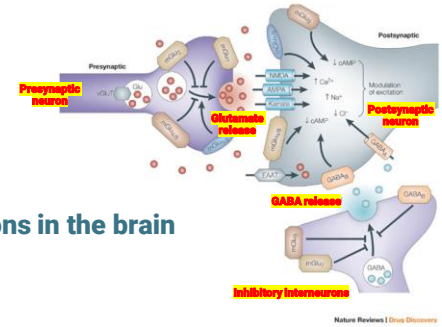
- Over 75% of the neurons in the brain are interneurons
- Especially important in the portion of the brain responsible for anxiety





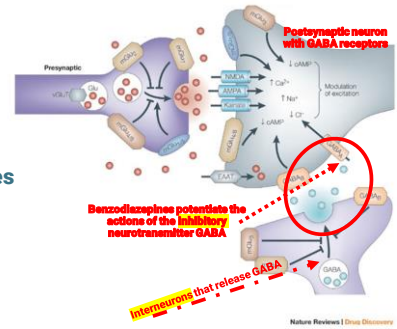
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Neurons in the brain



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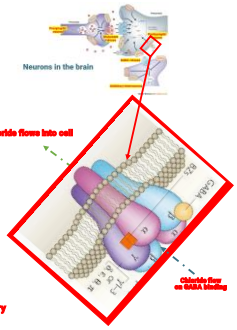
Benzodiazepines



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GABA

- GABA is the main inhibitory neurotransmitter – interneurons use GABA
- When GABA binds to receptor, chloride flows into the cell, making it more negative
  - This decreases the firing of those neurons
- Anxiety states are treated with drugs that potentiate the inhibitory actions of GABA or to open the GABA channels.



Opening chloride channels cancels the excitatory effect of opening sodium channels

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Benzodiazepines

- Benzodiazepines are the most commonly prescribed anxiolytics/hypnotics.
- Most of the benzodiazepines have the suffix *–ZEPAM or ZOLAM*
  - (*Diazepam* , *Alprazolam*, *Clonazepam* and *Lorazepam*)

Mechanism of action

- This group of medicines acts on gamma-aminobutyric acid (GABA) receptors in the CNS to potentiate its inhibitory action throughout the CNS

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Diazepam, Alprazolam and other ...zepam and ...zolam drugs

Diazepam first marketed as Valium

- Commonly used to treat anxiety, panic attacks, muscle spasms,
- Can be given parenterally
- LONG half life (50-100 hours)

Alprazolam is unique among the benzodiazepines, as it possesses antidepressant activity

- Addictive
- Half-life of 11.2 hours
- Caution in the elderly and those with hepatic disease
- Flumazenil (romazicon) is used in overdose

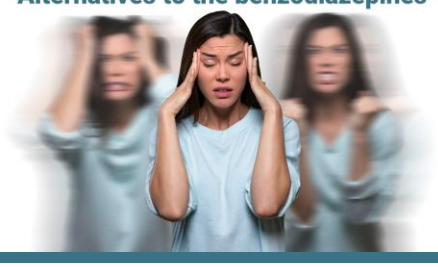
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Further information about benzodiazepines will be presented as we discuss the anti-seizure drugs



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Anxiolytics and hypnotics: Alternatives to the benzodiazepines



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Azapirones (*Buspirone*)

- Buspirone is an anxiolytic psychotropic drug
- It is primarily used to treat generalized anxiety disorder (GAD).
- It is not related to benzodiazepines or barbiturates
- No addiction potential
- Mechanism: serotonin receptor partial agonist
- Common adverse effects (>10%) Dizziness, Headache, sleepiness

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Other drugs with sedative and hypnotic effects

The FIRST GENERATION H1-antihistamines

- Available as over-the-counter sedative/hypnotic medicines.
  - *Hydroxyzine*
  - Half life in adults 20 hours; half life in children 7 hours
  - Drowsiness may remain the next day
  - Tolerance to the anti-anxiety and hypnotic effects may develop over time.

NOT suitable for elderly clients as the half-life increases to 30 hours and is associated with increased risk of developing Alzheimer's disease.

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Other drugs with sedative and hypnotic effects

α2-adrenoceptor agonists *Clonidine*

- Off-label treatment of anxiety
- Inhibits the release of norepinephrine (noradrenaline) in a form of negative feedback.
- **Also treats drug withdrawal (alcohol and opioids)**

Many antidepressants have anxiolytic effects
(see following section)

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Psychotropic medications: Part 3 Antidepressants



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Antidepressants: Definition and mechanisms of action

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Objectives for Part 3: CNS Antidepressants

- Briefly state the definition and most common theory of depression.
- Depression can be pharmaceutically treated using:
  1. **Selective serotonin reuptake inhibitors** (SSRI; *Sertraline*, *Escitalopram*, *Fluoxetine*).
  2. Serotonin noradrenaline reuptake inhibitors (SNRI's; *Duloxetine*, *Venlafaxine*)
  3. **Tricyclic antidepressants** (TCAs; *Amitriptyline*),
     - Also consider: Atypical (*Bupropion*, *Trazodone*)
- For each of the drug classes listed above, state one or more specific drugs that fall under that class, the general postulated mechanism by which it exerts its therapeutic effects, and the major side effects.

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Objectives for Part 3: Antidepressants

- Briefly state the definition and most common theory of depression.
- Depression can be pharmaceutically treated using:
  1. **Selective serotonin reuptake inhibitors** (SSRI; *Sertraline*, *Escitalopram*, *Fluoxetine*).
  2. Serotonin noradrenaline reuptake inhibitors (SNRI's; *Duloxetine*, *Venlafaxine*)
  3. **Tricyclic antidepressants** (TCAs; *Amitriptyline*),
     - Also consider: Atypical (*Bupropion*, *Trazodone*)
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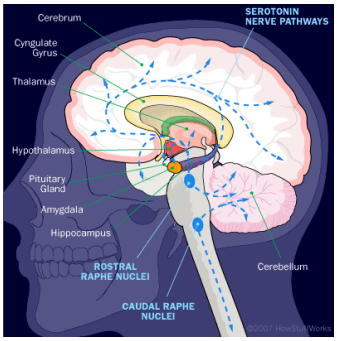
Depression and bipolar disease

- Depression is a state of profound sadness or melancholy. Common manifestations of this state include lethargy, apathy, loss of appetite, insomnia, feelings of unworthiness
  - Biogenic amine hypothesis does not explain everything
- In Bipolar disease, the patient experiences episodes of depression and mania, where the affected person cycles between the two extreme mood states.
- Mania/manic - abnormally elevated arousal energy level

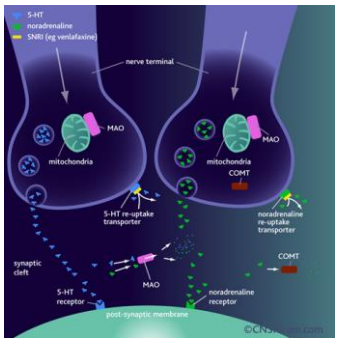
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DEPRESSION:

Abnormalities are commonly found in the lateral prefrontal cortex that is involved in the regulation of emotion

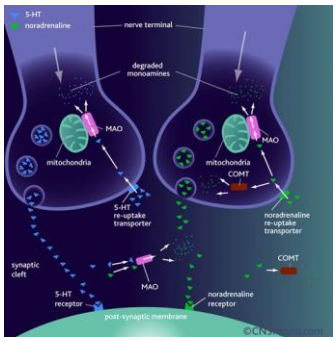


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In the areas of the brain associated with mood, we use serotonin and noradrenaline (NA)

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Monoamines are neurotransmitters that include:

- Serotonin (5HT),
- Dopamine,
- Noradrenaline (NA or NE for norepinephrine) and
- Adrenaline
- Histamine

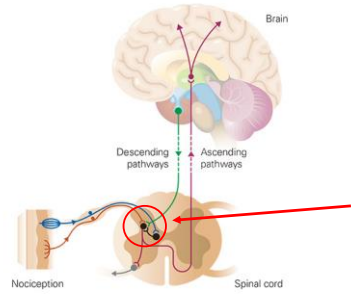
The deficit (or relative deficit) of serotonin or noradrenaline may be responsible for depression

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Monoamine oxidase inhibitors (MAOIs; phenelzine)

- **MAO is responsible for the degradation of the released noradrenaline.**
- **In people taking MAOIs, the ingestion of food rich in tyramine has the potential to precipitate a widespread release of noradrenaline in the periphery that is unable to be broken down because MAO is inhibited.**
- **Possible life-threatening cardiovascular stimulation.**

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Why are some antidepressants good for pain?

A: there are serotonin and noradrenaline receptors in the "gate" that can block incoming stimuli

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Antidepressants: SSRIs and TCAs

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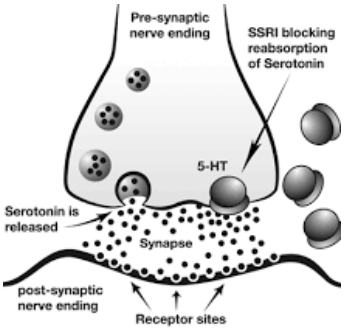
Selective serotonin reuptake inhibitors (SSRIs)

- ***Sertraline*** **(Zoloft) is the most widely used, followed by *Escitalopram (Lexapro)* and *Fluoxetine (Prozac)*.**
- **No common suffix** but most antidepressants end with -pram, -mine, -tine, -line
- **They are considered the first-line agents in the treatment of major depression.**
- **Take 4-6 weeks to be effective**
- **May increase risk of suicide especially in people under 18y/o**

PROZAC fluoxetine hydrochloride · serzone nefazodone HCl · zoloft (sertraline HCl) · Cymbalta duloxetine HCl · LUVOX fluvoxamine maleate · Lexapro escitalopram oxalate · PAXIL CR PAROXETINE HCl CONTROLLED-RELEASE TABLETS · VENLAFAXINE HCl EFFEXOR XR

The above depicts the most common SSRIs and SNRIs

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Selective serotonin reuptake inhibitors (SSRIs)

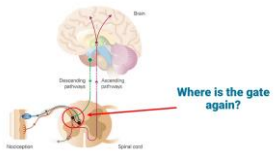
- ***Sertraline*** **(Zoloft) is the most widely used, followed by *Escitalopram (Lexapro)* and *Fluoxetine (Prozac)*.**
- **Many are indicated in major depression, panic disorder, social anxiety disorder, generalized anxiety disorder, and obsessive–compulsive disorder.**
- **Sexual dysfunction and decreased libido are among the most commonly reported side effects.**
- **Not to be withdrawn suddenly**

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Tricyclic antidepressants (TCAs)

TCAs are still important in the management of depression but are not always the antidepressants of choice.
Amitriptyline

- **Mechanism of action**
  - **These drugs act primarily by blocking the amine reuptake pumps on the presynaptic terminal in the brain.**
  - **The effect of decreasing pain occurs at 'the gate' of the spinal cord**



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Tricyclic antidepressants (TCAs)

- **Common adverse effects of TCAs**
  - Adverse effects derive from the fact that the TCAs have secondary actions:
  - Antimuscarinic actions: dry mouth, blurred vision, constipation, urinary retention and tachycardia**.
  - Central antihistamine actions: Mental confusion and sedation
  - Antiadrenergic activity: postural hypotension

Contraindicated in patients with history of cardiovascular problems or epilepsy
(remember they block reuptake of noradrenaline)

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SNRIs and Atypical Antidepressants

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SNRIs

Serotonin and noradrenaline reuptake inhibitors (SNRIs)

Mechanism of action

- Block the reuptake of noradrenaline, serotonin and, to some extent, dopamine from the synaptic gap. This action enhances neurotransmission
- *Duloxetine* (Cymbalta) and *Venlafaxine* (Effexor)



The above depicts the most common SSRIs and SNRIs

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SNRIs

Serotonin and noradrenaline reuptake inhibitors (SNRIs)

Mechanism of action

- Block the reuptake of noradrenaline, serotonin and, to some extent, dopamine from the synaptic gap. This action enhances neurotransmission

The SNRIs produce less sedation, as well as fewer cardiovascular and antimuscarinic side-effects compared to the TCAs.

- Always withdraw antidepressants slowly
- All antidepressants may increase risk of suicide

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Atypical antidepressants

***Bupropion* and *Trazodone* are atypical antidepressants.**

Bupropion is a weak noradrenaline and dopamine inhibitor

- Does not cause sexual dysfunction, weight gain or sleepiness
- Can be used as an add-on to SSRIs
- Is effective for reducing nicotine cravings

Trazodone blocks the reuptake of serotonin but it also blocks the postsynaptic serotonin receptors of subtype 2 A which are thought to contribute to depression.

- Indicated for depression, alcohol withdrawals and anxiety disorders
- Doesn't have anticholinergic effects

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Bipolar disorder (manic-depressive)

- Alternates between mania (euphoria) and depression
- Note that mania is the opposite to depression and is **not another term for extreme or acute**
- **Treatment:**
  - **Lithium carbonate**,
  - The antiseizure agents (**carbamazepine**, **sodium valproate**, **lamotrigine** and **clonazepam**)
  - Second generation antipsychotic agents **olanzapine**, **quetiapine**, **asenapine**, **risperidone** or **ziprasidone.**

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Lithium carbonate

- Lithium carbonate is the first-line therapy for the treatment and prevention of acute mania. It is a mood stabilizer.

Adverse effects

- Gastrointestinal irritation, tremor, muscle weakness and polyuria...
- Lithium has a **very narrow therapeutic range**. Toxicity results in exacerbation of above plus cardiac dysrhythmias, convulsions and renal failure.
- Plasma concentrations to be monitored closely

Drink plenty of water, check lithium and UEC levels frequently, do not consume other nephrotoxic agents

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Psychotropic medications: Part 4 Antipsychotics

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Objectives for Part 4: Antipsychotic agents

- Briefly explain the hypothesised neurologic abnormalities that are found in individuals with schizophrenia and other forms of psychosis.
- Explain the four major pathways for dopamine 2 in the brain and relate those to the actions of our antipsychotic medications
- For each of the drugs noted below, be able to identify the drug class to which it belongs. (Simply identify it as a typical first class antipsychotic or atypical.)
  - Typical antipsychotic agents include Haloperidol and Chlorpromazine, and
  - Atypical antipsychotic agents include *Quetiapine*, *Aripiprazole*, *Risperidone*, and *Olanzapine*
- For both of the drug classes above, briefly describe the mechanism by which drugs of that broad class exert their therapeutic effect, the general indications, contraindications and major side effects.

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Antipsychotic agents

- **Psychoses:** mental health disorders characterised by aberrant behaviour and disturbed emotional states.
  - Schizophrenia, severe agitation and some forms of dementia
- The most effective medicines in the treatment of the psychoses are the **central dopamine receptor antagonists**



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Schizophrenia

- **Schizophrenia is relatively common, affecting 1 per cent of the population,**
  - **Increased dopaminergic activity in certain parts of the brain, particularly the prefrontal cortex, limbic**
  - **Onset 15 and 25**

Symptoms of schizophrenia are described as either negative or positive.

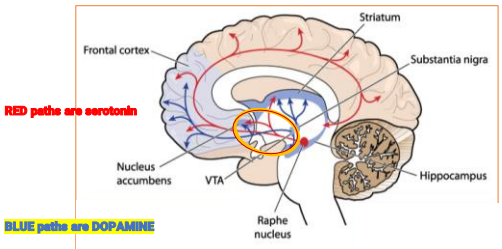
- **Positive symptoms**
  - **delusions, disorganised speech and/or thought, distorted perceptions and emotions, hallucinations and the belief that one is under the control of an external agency.**
- **Negative symptoms**
  - **toneless speech, vacant eyes, social withdrawal, anhedonia (lack of enjoyment) and apathy.**

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Dopamine pathways

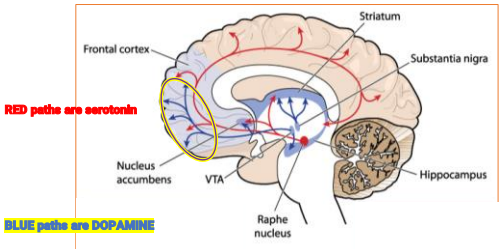
- Mesolimbic pathway – hyperactive in schizophrenia – yields positive symptoms like delusions
- Mesocortical pathway - underactive in schizophrenia – contributes to negative symptoms
- Nigrostriatal pathway – decreases in dopamine here result in Parkinsonian symptoms
- Tuberoinfundibular pathway – controls prolactin release

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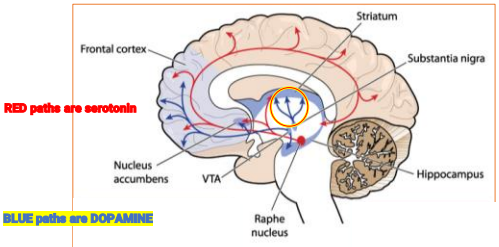
Mesolimbic pathway – hyperactive in schizophrenia – yields positive symptoms like delusions

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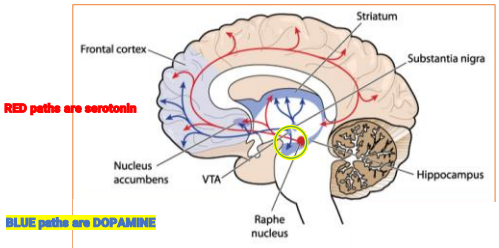
Mesocortical pathway - underactive in schizophrenia – contributes to negative symptoms

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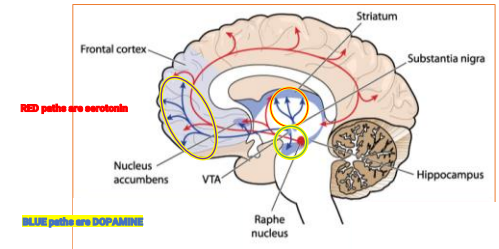
Nigrostriatal pathway – decreases in dopamine here result in Parkinsonian symptoms

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Tuberoinfundibular pathway – controls prolactin release

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Mesocortical pathway - underactive in schizophrenia – contributes to negative symptoms

Nigrostriatal pathway – decreases in dopamine here result in Parkinsonian symptoms

Tuberoinfundibular pathway – controls prolactin release

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First generation antipsychotics

- First generation antipsychotics
  - Also termed 'typical' 'neuroleptics' or 'major tranquillisers'
- Haloperidol (Serenace) and Chlorpromazine (Largactil)
- Similar mechanisms of action on the CNS
- Main action is to antagonise dopamine receptors
  - Numerous adverse effects

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Adverse effects of Haloperidol, Chlorpromazine and other first generation antipsychotic agents

- Worsening of the negative symptoms (lack of motivation, diminished energy, inability to express emotions, social withdrawal)
- Parkinsonian symptoms
- Excess prolactin (milk production, breast development in males and sexual dysfunction)
- Long term effects include the opposite of parkinsonian symptoms, that being tardive dyskinesia (smacking of lips, etc)

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Additional adverse effects of first generation antipsychotic agents

High potency agents

Example: Haloperidol

Cause more profound extrapyramidal symptoms and more galactorrhoea

Low potency agents

Example: chlorpromazine

- Bind to alpha adrenergic receptors
  - cause orthostatic hypotension
- Bind to the acetylcholine receptor
  - Cause eye pain
  - Dry mouth
  - Constipation
- Bind to histamine 1 receptor
  - Drowsiness,
  - Sedation,
  - Weight gain

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Second generation antipsychotics

Second-generation (atypical) antipsychotics

- *Quetiapine* (Seroquel), *Aripiprazole* (Abilify), *Risperidone* (Risperdal), and *Olanzapine* (Zyperxa)
- The most important difference between the first and second generation antipsychotics is the tendency of the latter to induce fewer extrapyramidal disturbances.
- Differences in their affinities for various central receptors may explain why the second generation antipsychotics are less prone to producing these adverse effects.

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Second generation antipsychotic agents

- *Quetiapine*
  - Binds to Histamine 1 receptors and cause drowsiness
- *Aripiprazole*
- *Olanzapine*
  - Binds to Serotonin 2c strongly and causes metabolic symptoms such as weight gain and high glucose
  - Bind to Histamine 1 receptors and cause drowsiness
- *Risperidone*
  - Binds strongly to alpha 1 receptors (orthostatic hypotension)
  - Highest amount of extrapyramidal symptoms and galactorrhoea

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First generation vs second generation antipsychotics: review

- Antipsychotic agents can be classified as either typical (first generation) or atypical agents (second generation).
- Typical are much more active on the D2 receptors than are the atypical ones.
- The atypical agents, have less affinity for the D2 receptors, usually have a lower incidence of debilitating motor disturbances

The goal is to find the right antipsychotic agent for the person: an agent that addresses the most disturbing symptoms without the side effects that may prompt the person to stop taking the medication.

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Southern Cross University

Psychotropic medications: Part 5
Antiseizure medications

Dr Ken Wojcikowski

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Anti-seizure medications



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Objectives for Part 5
Anti-seizure Medicines

- Define the terms seizure, partial or focal seizure, generalized seizure, epilepsy, and status epilepticus.
- For each of the following drugs, briefly explain the mechanism of action, and the major indications and side effects.
  - Clonazepam and diazepam (the benzodiazepines)
  - Gabapentin and Pregabalin,
  - Carbamazepine and valproic acid (valproate)

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Terms regarding seizures*

*The term anticonvulsant is misleading because some seizures do not manifest as convulsions.

- Partial seizures – begin in one hemisphere at a single cortical focus
- Generalised seizures form across both hemispheres
  - Consciousness is always lost
- Epilepsy is a term used to indicate a condition of recurrent seizures
  - It does not describe the nature of the seizure itself.
- Status epilepticus: the affected person moves immediately from one seizure into another.
  - Very dangerous unless treated immediately

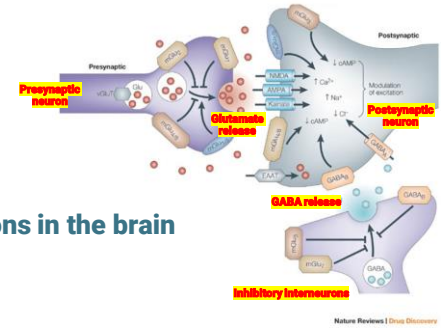
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How do you treat or prevent a seizure?

1. Increase chloride anions (the negative charge will decrease brain over activity)
2. Decrease sodium ions (the decrease in positive ions will decrease brain over activity)
3. Decrease calcium ions (the decrease in positive ions will decrease brain over activity)

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Neurons in the brain

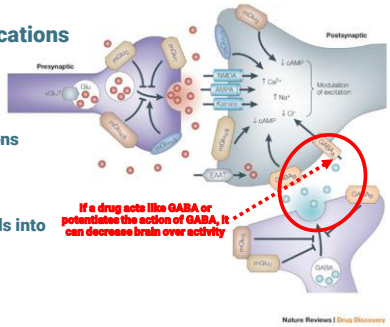


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Antiseizure medications

General mechanisms

- Inhibitory interneurons release GABA
- Gaba binds to GABA receptor
- Chloride anion travels into the cell
- More inhibition

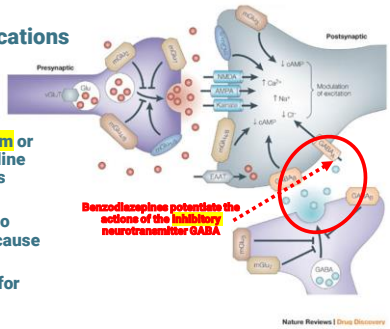


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Antiseizure medications

Benzodiazepines

- Intravenous diazepam or lorazepam are first-line treatments for status epilepticus
- Should not be used to prevent seizures because tolerance develops
- (refer to anxiolytics for further information)

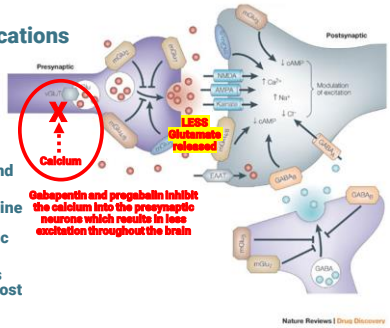


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Antiseizure medications

Gabapentin and Pregabalin

- Decrease activity of calcium channels
- Treat partial seizures and neuropathic pain
- Gabapentin is the first-line medication for the treatment of neuropathic pain
- Side effects: sleepiness and dizziness are the most common.

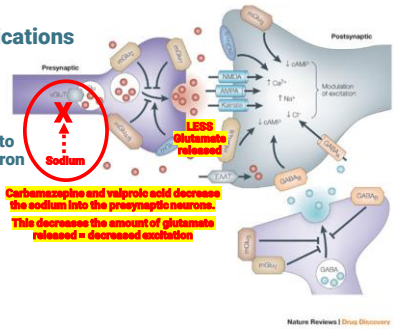


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Antiseizure medications

Carbamazepine and valproic acid

- Decrease sodium into the presynaptic neuron



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Carbamazepine

Carbamazepine (CBZ) is an anti seizure and mood-stabilizing drug used primarily in the treatment of epilepsy and neuropathic pain.

- Generalized seizures

CBZ is a sodium channel blocker .

Adverse effects may include nausea and drowsiness

Bone marrow dysfunction is possible

- Check LFTs and FBCs periodically!

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Valproic acid

Valproic acid is an anti seizure and mood-stabilizing drug used primarily in the treatment of epilepsy and bipolar disorder.

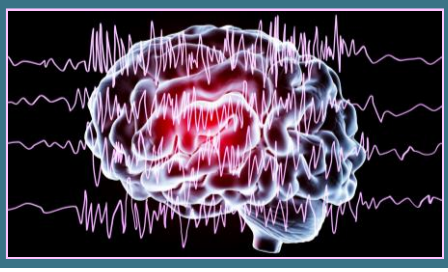
Partial and generalized seizures

Valproic acid is a sodium channel blocker (and may increase GABA).

Side effects include nausea, sleepiness, dry mouth

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Anti-seizure medications



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